KlearTrust Response to CMS Request for Information (CMS-0042-NC)
Health Technology Ecosystem

Value is not a retrospective metric or a static rebate; it is the real-time realization of clinical effectiveness and economic return, shared transparently across all stakeholders.

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KlearTrust Intro

KlearTrust's unique service model enables scalable, data-driven, and transparent administration of value-based healthcare programs. It delivers a unified infrastructure for the design, execution, and reconciliation of innovative contracting strategies between life sciences manufacturers, payers, and providers.

Value is not a retrospective metric or a static rebate; it is the real-time realization of clinical effectiveness and economic return, shared transparently across all stakeholders.

More precisely, value under KlearTrust reflects the intersection of:

1. Clinical Impact
    - Achievement of desired therapeutic outcomes (e.g., remission rates, reduced hospitalizations, improved adherence).
    - Timeliness and appropriateness of care, particularly in high-stakes disease states (e.g., oncology, rare disease, cardiometabolic conditions).
2. Economic Efficiency
    - Reduction in total cost of care through appropriate therapy utilization, fewer avoidable complications, and improved resource allocation.
    - Risk-sharing between stakeholders (manufacturers, payers, providers) based on how well a therapy or intervention performs in the real world.
3. Operational Transparency & Trust
    - Accurate, event-driven tracking of therapy initiation, progression, discontinuation, and outcomes—enabled by KlearTrust's Continuous Active Intelligence (C.A.I.).
    - Automated, unbiased reconciliation of value-based contracts using integrated claims, EMR, lab, and pharmacy data.

Executive Summary

KlearTrust appreciates the opportunity to respond to the Centers for Medicare & Medicaid Services (CMS) and the Office of the National Coordinator for Health Information Technology (ONC) regarding the Request for Information (CMS–0042–NC) on the Health Technology Ecosystem. We commend CMS's leadership in advancing a future-ready, patient-centric digital infrastructure for healthcare. In our view, a critical enabler of this future is what we call Continuous Active Intelligence (CAI)—a capability that transforms passive health data into real-time, proactive, and contextualized insights across the care and payment continuum.

What Is Continuous Active Intelligence (CAI)?

CAI is a technology and data operations paradigm that goes beyond retrospective analytics and static reporting. It refers to the continuous ingestion, harmonization, and real-time analysis of multi-source healthcare data—including claims, clinical, social, behavioral, and administrative data—to create dynamic situational awareness and decision support.

CAI systems leverage modern data standards like HL7 FHIR, event-driven architectures, and machine intelligence to produce persistent, high-frequency insights. These insights are not just

descriptive but causal, predictive, and actionable—guiding stakeholders toward early, efficient interventions and continuous alignment with value-based goals.

Why CAI Is Essential

Today's healthcare environment remains burdened by fragmented data, lagging analytics, and administrative bottlenecks. These limitations result in:

- Delayed interventions and missed care opportunities
- Limited visibility into evolving patient risk
- Contract performance blind spots
- Onerous administrative overhead, especially in prior authorization and documentation workflows

CAI addresses these challenges by transforming interoperability from a technical exchange of data into a functional exchange of intelligence—available at the point of care, at the point of adjudication, and at every decision point in between.

What Value Can CAI Deliver?

Implemented at scale, CAI delivers tangible benefits to all stakeholders:

Patients: Timelier, better-coordinated care; digital quality transparency; personalized interventions

Providers: Reduced administrative burden; real-time quality tracking; intelligent decision support

Payers: Improved financial and administrative outcomes through faster, evidence-based prior authorization and adjudication; prospective risk management; fraud/waste detection; Value-Based care contract administration insights and outcomes measurement; provider benchmarking insights; clinical pathway adherence insights

Manufacturers: Enhanced outcomes visibility; value-based contract alignment; early signal monitoring

CMS & Policymakers: Accelerated APM impact; continuous oversight of quality and equity metrics; scalable policy feedback loop

By integrating CAI into the digital backbone of healthcare, CMS can empower the transition from a reactive to a proactive system—one that is intelligent by default, equitable by design, and scalable by standards.

The Role of FHIR in Enabling CAI

HL7 FHIR provides the ideal technical foundation for CAI:

- RESTful APIs for real-time data access and exchange
- Subscriptions and notifications for event-driven triggers
- Atomic transactions and bundles to preserve consistency and context
- SMART on FHIR integration for secure, EHR-embedded applications

These capabilities allow CAI to operate across the ecosystem, unlocking a new class of real-time, standards-aligned applications that reduce friction, increase responsiveness, and align outcomes to value.

Our Commitment

KlearTrust is purpose-built to deliver CAI. Our platform ingests and analyzes real-time clinical, claims, and patient-reported data using FHIR APIs, supporting smarter care, faster approvals, and transparent accountability. We urge CMS to incorporate FHIR-enabled CAI capabilities into the core fabric of interoperability programs, certification criteria, and value-based payment infrastructure.

We thank CMS for the opportunity to contribute to this vital dialogue and welcome the opportunity to collaborate further.

Key Themes in Response to the RFI

Real-Time Data Collection and Surveillance for CAI

KlearTrust proposes a paradigm shift from retrospective analytics to real-time, proactive surveillance of the patient journey. This includes continuous ingestion, normalization, and analytics across multi-source data including:

- Medicare claims (Parts A, B, D)
- Clinical data from certified EHRs including unstructured notes and documents
- Social determinants of health
- Patient-reported outcomes
- Pharmacy data
- Diagnostic Data
- Utilization trends and care coordination records
- Admit and discharge data

This integrated surveillance delivers Continuous Active Intelligence (CAI) that:
- Alerts stakeholders to care gaps and non-adherence
- Detects preventable admissions and readmissions
- Identifies early opportunities for intervention
- Enhances alignment with quality and contract performance metrics

Workflow Optimization and Prior Authorization Automation

Administrative inefficiencies such as prior authorization represent significant friction points in healthcare. Rather than challenging the necessity of PA policies, KlearTrust targets automation of

documentation delivery:

- Identifying relevant data elements for specific authorization requests
- Extracting structured and unstructured clinical content (labs, imaging, notes)
- Mapping to payer-specific requirements using standards-based APIs
- Delivering payer-ready packages in real time

This targeted automation significantly reduces provider administrative burden while accelerating patient access to needed care.

API Enablement and Interoperability Standards

KlearTrust strongly supports CMS's continued investment in API-first interoperability. We advocate for:

- Expanding FHIR endpoints to support real-time data streaming and alerting
- Including SDoH, care management, and remote monitoring data within API frameworks
- Promoting a unified digital identity model for patients and providers
- Establishing real-time data validation and provenance standards

This will unlock the potential of third-party applications that deliver CAI through safe, trusted, and dynamic data flow.

Supporting Value-Based Care and Alternative Payment Models

CAI is essential to scaling and sustaining value-based care. It supports:

- Real-time attribution monitoring
- Risk stratification and patient cohort identification
- Measurement of clinical and contractual compliance
- Early identification of financial risk leakage

We encourage CMS to incorporate CAI-enabling capabilities into CEHRT definitions and APM technology requirements. Targeted funding for adoption in rural or resource-constrained settings would also support equity and access.

Point-by-Point Responses to RFI Questions from the CAI Perspective

2. Data Access and Integration (PC–8 to PC–12)

PC–8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

From the perspective of Continuous Active Intelligence (CAI), claims data (Parts A, B, and D), basic demographics, and problem lists from certified EHRs are among the most readily available data. However, their utility is often constrained by delay, inconsistency, and lack of context. To be most valuable, data must be both real-time and contextualized.

Valuable but hard-to-access data includes unstructured EHR notes, imaging and pathology reports, patient-generated health data, and social determinants of health (SDoH) indicators.

These are crucial for CAI systems to deliver timely, context-sensitive insights. Barriers include lack of FHIR representation, inconsistent coding, and minimal patient-level integration across systems.

PC–9. Blue Button 2.0 API expansion recommendations

Developers view the following CMS data sources as high-value additions to Blue Button 2.0:

- Encounter notes and care summaries
- Prior authorization decisions
- Real-time formulary and benefit information
- Functional status assessments and PROMs

Current barriers include delayed claims availability and lack of structured outputs for unstructured fields. Improving documentation standards and expanding the API to include adjudicated and unaudited claim drafts, would support CAI applications in surfacing early indicators of utilization or non-adherence.

PC–10. TEFCA Usefulness and Recommendations

TEFCA offers foundational promise in supporting trust, identity management, and universal exchange protocols. However, CAI requires TEFCA to accelerate real-time, event-driven exchange. Use cases with high CAI impact include:
- Automated updates to care plans across care teams
- Triggered alerts for new diagnoses or care transitions
- Standardized push notifications for lab result thresholds

We recommend prioritizing use cases that support prospective surveillance and developing payload standards for CAI-compatible exchange.

PC–11. Role of Health Information Exchanges (HIEs)

High-performing HIEs can be critical partners in CAI. Their value increases when data quality, completeness, and timeliness meet the thresholds needed for continuous monitoring. Successful models demonstrate:

- Integration of hospital and ambulatory feeds
- Use of advanced patient matching algorithms
- FHIR and ADT-based exchange mechanisms

CMS should consider grants or incentives for HIEs that achieve CAI-readiness criteria such as data latency thresholds and semantic normalization.

PC–12. Valuable Operational Use Cases for CAI

KlearTrust identifies several use cases that offer outsized value for CAI-enabled operational improvements:

- Real-time provider directory with availability scheduling
- Patient-specific cost prediction tools linked to authorization and eligibility
- Patient-facing digital quality performance dashboards
- Third-party CAI apps that manage and update dynamic care plans

While some of these are technically possible, data silos and variable access policies hinder widespread adoption. CMS leadership in standardizing access mechanisms will accelerate innovation.

Section: Value-Based Care Organizations – Digital Health Adoption (VB–1 to VB–4)

VB–1. Incentives and Obstacles for APMs to Adopt Digital Health Tools

CAI enables APMs and ACOs to proactively monitor patient risk, optimize quality metrics, and close care gaps before they affect performance. To encourage adoption of CAI-powered digital health tools:

Recommended Incentives:
- Shared savings bonus adjustments for organizations that integrate CAI tools with certified EHRs and demonstrate real-time data exchange and proactive patient engagement.
- Add-on payments or infrastructure grants for providers implementing standards-aligned CAI workflows for care navigation, particularly in underserved areas.
- Performance measurement relief for APM entities that deploy CAI capabilities to track real-time attribution, clinical event detection, and outcomes forecasting.

Key Barriers:
- Fragmented and delayed data flows
- Lack of dedicated funding for CAI tool integration
- Uncertainty over whether CAI-enabled tools qualify under CEHRT definitions
- Misalignment between current quality measurement timelines and the real-time analytics CAI enables

VB–2. Integrating Emerging Technologies into APM Requirements

CMS should explicitly align APM program requirements with CAI-related capabilities such as:

- Real-time population health surveillance via FHIR-enabled data aggregation
- Embedded AI algorithms that detect clinical deterioration or non-adherence before adverse events
- Live care coordination dashboards populated by streaming patient journey data
- Integrated quality tracking tools that run continuously rather than retrospectively

Additionally, CAI should be recognized as a foundational capability for supporting:

- Dynamic patient segmentation
- Intelligent attribution adjustment
- Continuous feedback loops for quality performance and operational compliance

VB–3. Essential Health IT Capabilities for Value-Based Care

From a CAI perspective, the following are essential health IT capabilities to support value-based care:
- Event-driven interoperability: Use of FHIR subscriptions and ADT feeds to detect and respond to patient events in real time
- Longitudinal patient surveillance: Continuous ingestion of clinical, claims, SDoH, and patient-reported data to maintain active patient context
- Predictive analytics and intelligent triggers: AI models surfacing actionable insights for care gaps, risk shifts, or guideline deviations
- Real-time quality measurement engines: Enabling on-demand compliance tracking at the patient and population level
- Smart documentation automation: Reducing the burden of prior authorization and risk adjustment submissions through CAI-based clinical data extraction

VB–4. Essential Data Types for VBC Success

CAI systems depend on a high-resolution, multi-dimensional view of the patient. The following data types are essential:

- Structured and unstructured clinical data (notes, labs, diagnostics, vitals, care plans)
- Claims and utilization data (encounters, procedures, prescriptions, adjudicated and pending)
- Social determinants of health (housing, food, transportation insecurity, community factors)
- Patient-reported outcomes and adherence data
- Real-time status feeds (ADT notifications, care transitions, referrals)

CMS can support broader access by requiring timely FHIR-based exchange of these datasets across all certified technology platforms, and incentivizing providers to activate this data within CAI-enabled tools.